Machine Learning

AIML CZG565

M4 : Linear Models for Classification

BITS Pilani
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Course Faculty of MTech Cluster

BITS – CSIS - WILP

Disclaimer and Acknowledgement

- These content of modules & context under topics are planned by the course owner Dr. Sugata, with grateful acknowledgement to many others who made their course materials freely available online
- We here by acknowledge all the contributors for their material and inputs.
- We have provided source information wherever necessary
- Students are requested to refer to the textbook w.r.t detailed content of the presentation deck shared over canvas
- We have reduced the slides from canvas and modified the content flow to suit the requirements of the course and for ease of class presentation

Slide Source / Preparation / Review:

From BITS Pilani WILP**:** Prof.Sugata, Prof.Chetana, Prof.Rajavadhana, Prof.Monali, Prof.Sangeetha, Prof.Swarna, Prof.Pankaj

External: CS109 and CS229 Stanford lecture notes, Dr.Andrew NG and many others who made their course materials freely available online

Recap

- Classification
  - Output is discrete e.g. f(x) : {yes, no, maybe}
  - f(x) : {0, 1} or {yes, no} Binary Classification
  - Ojective: minimize mis-classification rate

- Two kinds of Classifiers:
  - Probabilistic Generative Classifiers
    - Models aim to learn joint probability distribution P(X, Y)
  - Probabilistic Discriminative Classifiers
    - Models learn the decision boundary that separate the two classes

AGENDA

Logistic Regression

Logistic Regression

- statistical model used for binary classification problems,
  - goal is to predict the probability that a given input belongs to one of two classes.
  - uses the logistic function to model the relationship between the input variables (also known as predictors or features) and the output variable (also known as the response or target)

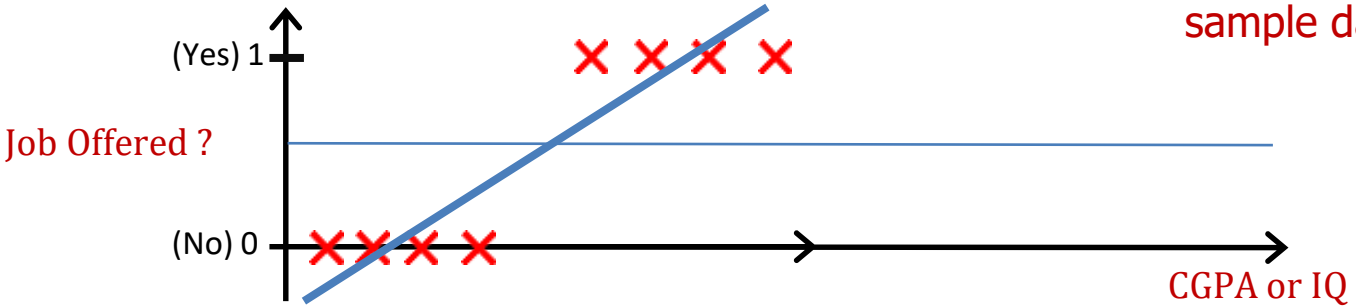
    MODEL:

    z = woxo + w1 x1 + …. + wnxn

    hz(x) = 1/ 1 + e^ -z

Logistic Regression vs Least Squares Regression

Figure depicts sample data points



- Independent Attribute : CGPA or IQ
- Can we solve the problem using linear regression? E.g., fit a straight line and define a threshold at 0.5
- Threshold classifier output $h_\theta(x)$ at 0.5:

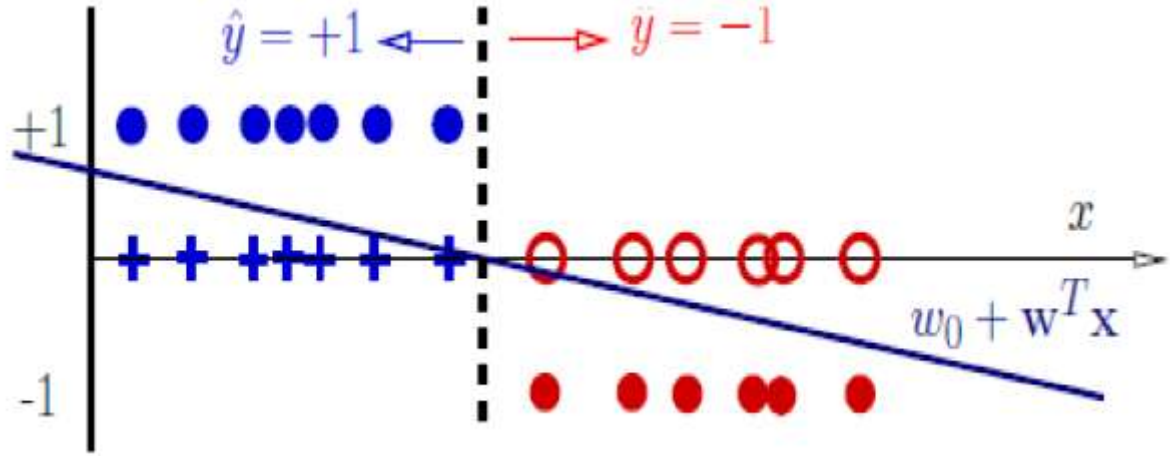
A Discriminant function f (x) directly map input to class labels
In two-class problem, f (.) is binary valued

If $h_\theta(x) \geq 0.5$, predict "y = 1"

If $h_\theta(x) < 0.5$, predict "y = 0"

In this use case : $h_\theta(x) = 0.7$, implies that there is 70% of chance of the candidate being selected in the interview

Decision Rules



- Classifier:

  $f(\mathbf{x}, \mathbf{w}) = w_o + \mathbf{w}^T\mathbf{x}$ (linear discriminant function)

- Decision rule is

$$y = \begin{cases} 1 & \text{if } f(\mathbf{x}, \mathbf{w}) \geq 0 \\ -1 & \text{otherwise} \end{cases}$$

- Mathematically

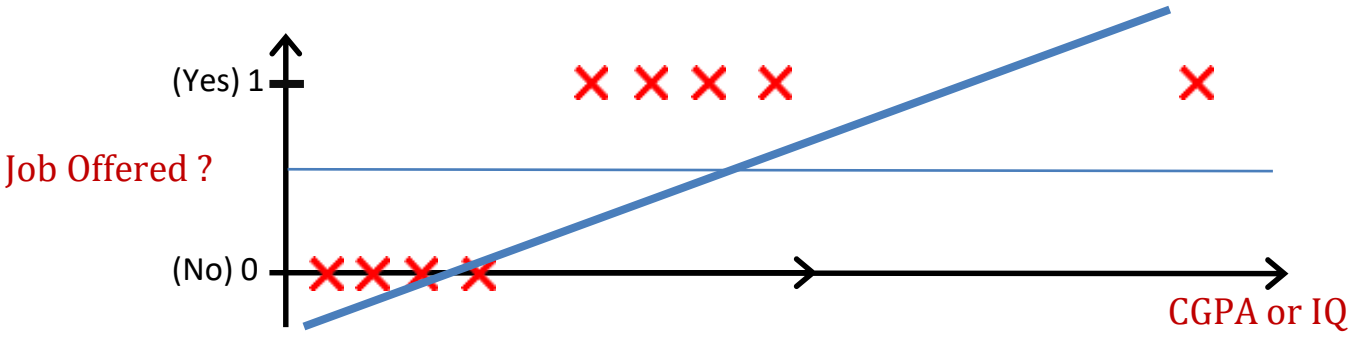
  $y = \text{sign}(w_0 + \mathbf{w}^T\mathbf{x})$

- This specifies a linear classifier: it has a linear boundary (hyperplane)

  $w_0 + \mathbf{w}^T\mathbf{x} = 0$

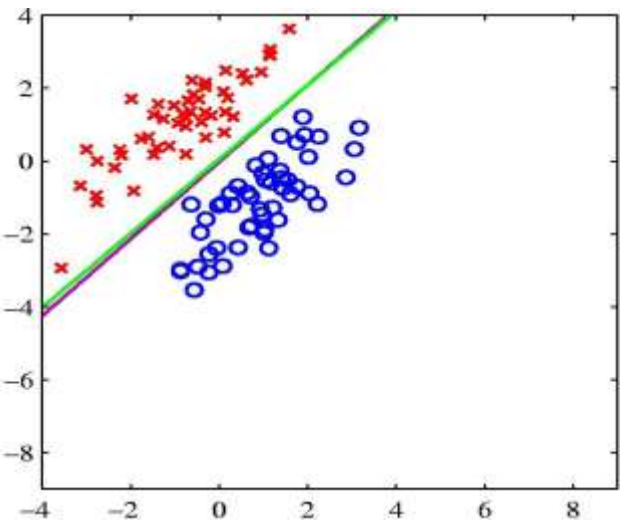
A discriminant is a function that takes an input vector x and assigns it to one of K classes, denoted C_k.

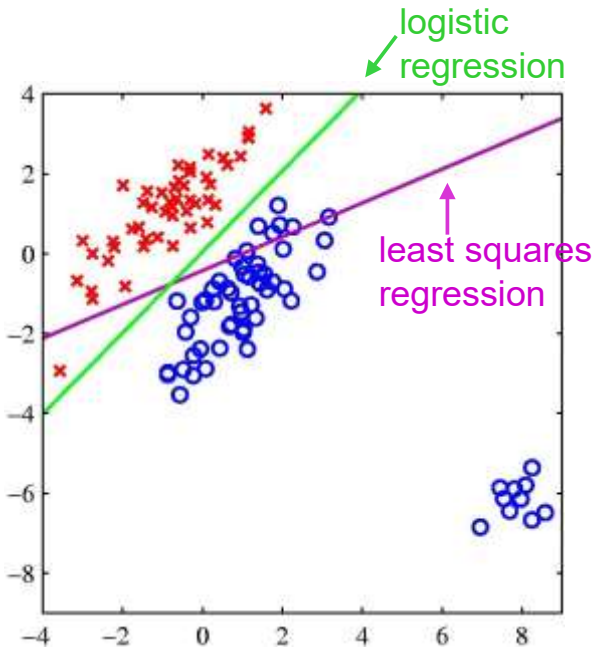
Logistic Regression vs Least Squares Regression



Failure due to adding a new point

Logistic Regression vs Least Squares Regression





The right-hand plot shows the corresponding results obtained when extra data points are added at the bottom left of the diagram, showing that least squares is highly sensitive to outliers, unlike logistic regression.

Sigmoid Function

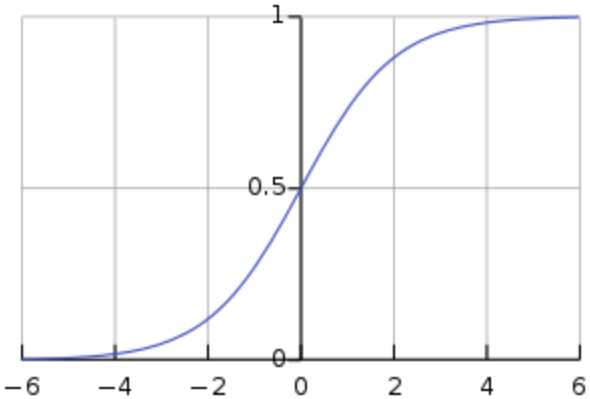
- Sigmoid/logistic function takes a real value as input and outputs another value between 0 and 1
- That framework is called logistic regression
  - Logistic: A special mathematical sigmoid function it uses
  - Regression: Combines a weight vector with observations to create an answer

- Want $0 \leq h_\theta(x) \leq 1$

- $h_\theta(x) = g(\theta^\top x)$,

  where $g(z) = \frac{1}{1+e^{-z}}$

- Sigmoid function
- Logistic function



$$h_\theta(x) = g(\theta^T x)$$

$$\boxed{h_\theta(x) = \frac{1}{1+e^{-\theta^\top x}}}$$

Logistic Regression vs Least Squares Regression

Classification: y = 0 or 1

Another drawback of using linear regression for this problem

$$Y = \theta_0 + \sum_i \theta_i x_i$$

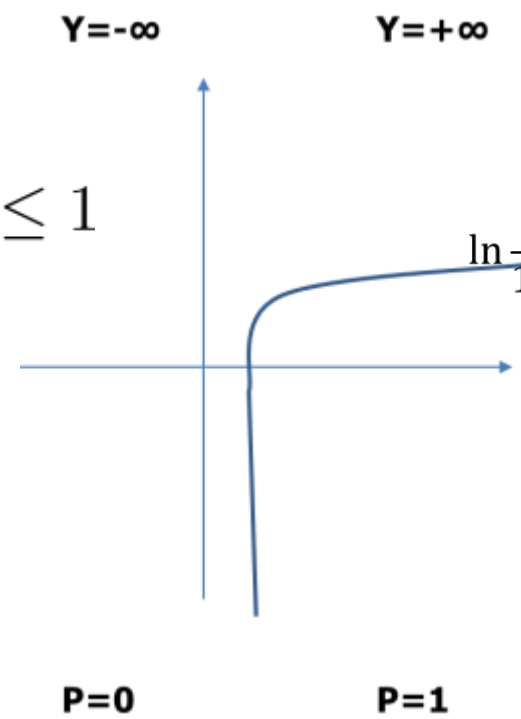
$h_\theta(x)$ can be > 1 or < 0

$$p = \frac{1}{1 + e^{-y}}$$

$$h_\theta(x) = g(\theta^T x)$$

What we need: $0 \leq h_\theta(x) \leq 1$

$$\ln \frac{p}{1-p} = \theta_0 + \sum_i \theta_i x_i$$



Logistic Regression – Sample Linear Boundary

- At decision boundary output of logistic regression is 0.5
- $h_\theta(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$
  - e.g., $\theta_0 = -3$, $\theta_1 = 1$, $\theta_2 = 1$

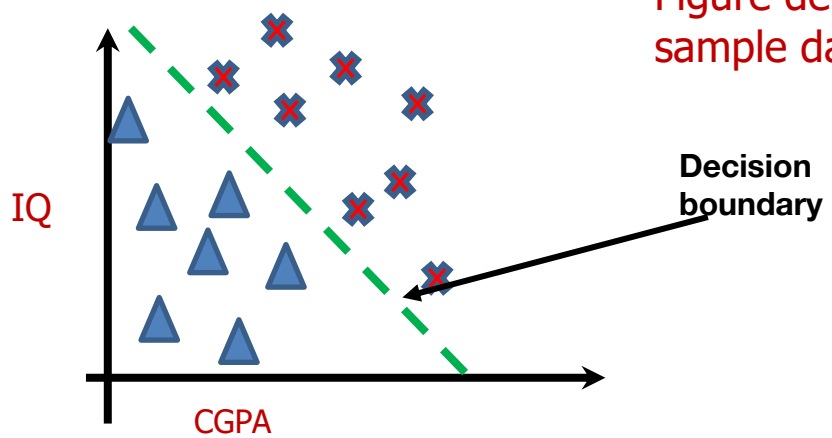
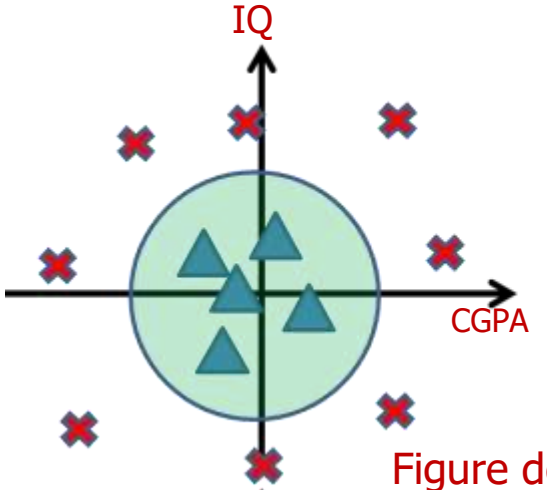


Figure depicts sample data points

- Predict "$y = 1$" if $-3 + x_1 + x_2 \geq 0$

Slide credit: Andrew Ng

Logistic Regression – Sample Non-Linear Boundary

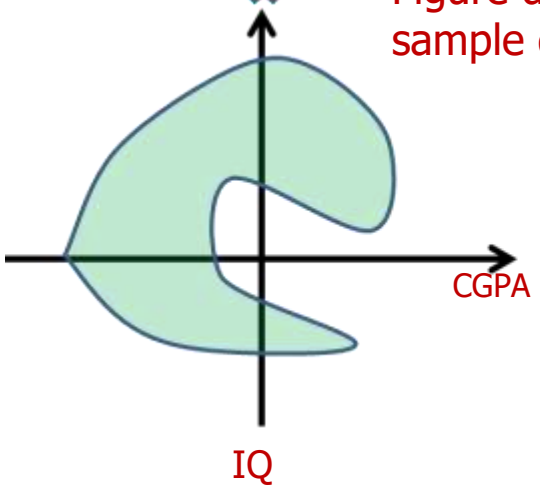


- $h_\theta(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_2^2)$

E.g., $\theta_0 = -1, \theta_1 = 0, \theta_2 = 0, \theta_3 = 1, \theta_4 = 1$

- Predict "$y = 1$" if $-1 + x_1^2 + x_2^2 \geq 0$

Figure depicts sample data points



- $h_\theta(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_1^2 x_2 + \theta_5 x_1^2 x_2^2 + \theta_6 x_1^3 x_2 + \cdots)$

Slide credit: Andrew Ng

Learning model parameters

- Training set:

$$\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \cdots, (x^{(m)}, y^{(m)})\}$$

- m examples

$$x \in \begin{bmatrix} x_0 \\ x_1 \\ \cdots \\ x_n \end{bmatrix} \qquad x_0 = 1, y \in \{0, 1\}$$

$$h_\theta(x) = \frac{1}{1 + e^{-\theta^T x}}$$

- **How to choose parameters (feature weights) ?**

Notion of Cost Function in Classification

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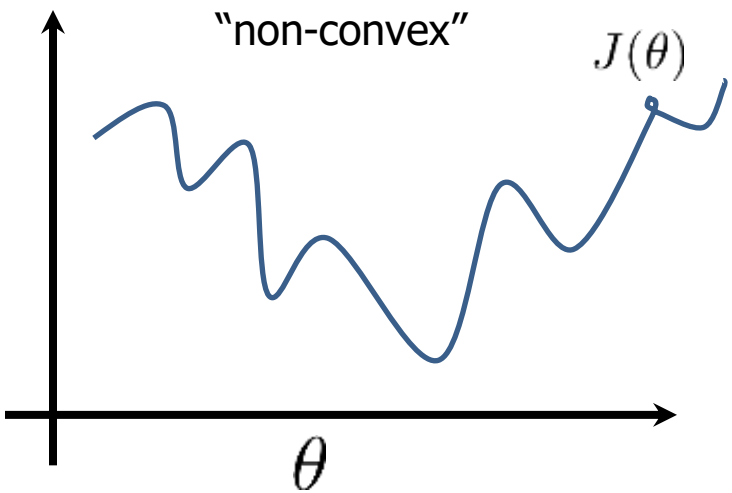
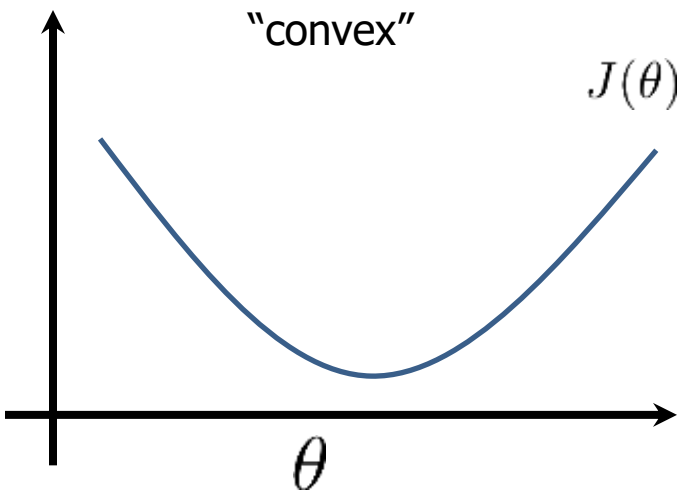
Logistic Regression

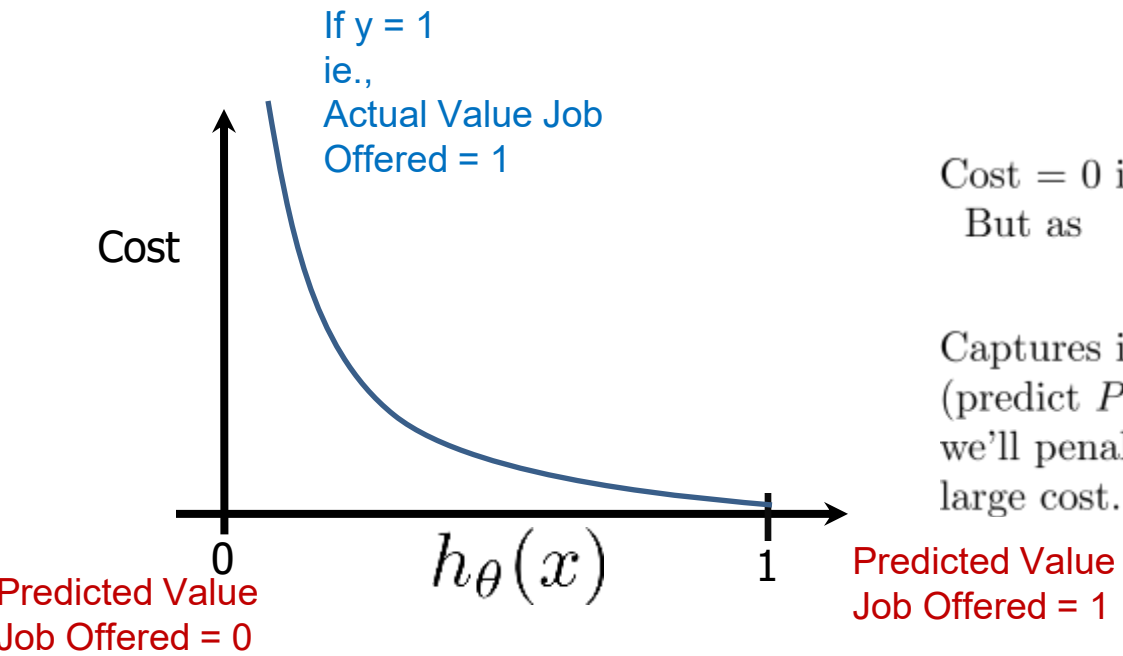
- Training set: $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \cdots, (x^{(m)}, y^{(m)})\}$
- How to choose parameters (feature weights)? θ





Logistic regression cost function (cross entropy)

$$\mathrm{Cost}(h_\theta(x), y) = \begin{cases} -\log(h_\theta(x)) & \text{if } y = 1 \\ -\log(1 - h_\theta(x)) & \text{if } y = 0 \end{cases}$$



Cost $= 0$ if $y = 1, h_\theta(x) = 1$

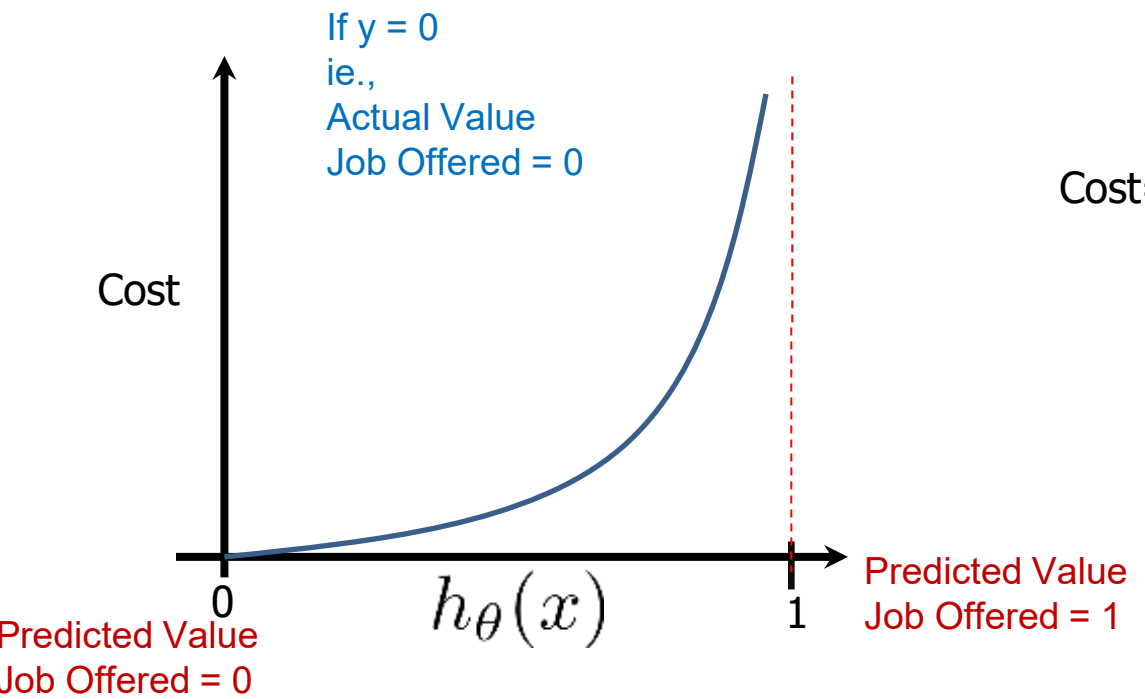
But as $\quad h_\theta(x) \to 0$

$\quad\quad\quad\quad Cost \to \infty$

Captures intuition that if $h_\theta(x) = 0$, (predict $P(y = 1|x; \theta) = 0$), but $y = 1$, we'll penalize learning algorithm by a very large cost.

Logistic regression cost function

$$\mathrm{Cost}(h_\theta(x), y) = \begin{cases} -\log(h_\theta(x)) & \text{if } y = 1 \\ -\log(1 - h_\theta(x)) & \text{if } y = 0 \end{cases}$$



Cost=0; If y=0 and $h_\theta(x)=0$

Cost function

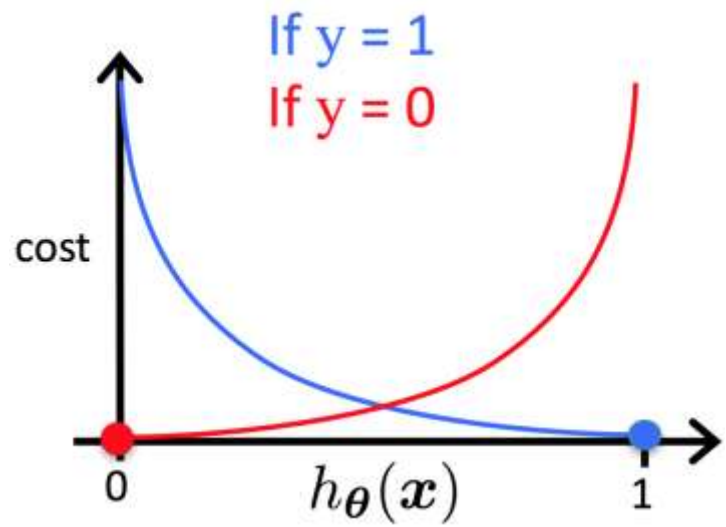
$$J(\theta) = \frac{1}{m}\sum_{i=1}^{m} \mathrm{Cost}(h_\theta(x^{(i)}), y^{(i)})$$

$$= -\frac{1}{m}[\sum_{i=1}^{m} y^{(i)} \log h_\theta(x^{(i)}) + (1 - y^{(i)}) \log (1 - h_\theta(x^{(i)}))]$$

To fit parameters θ : Apply Gradient Descent Algorithm $\min_\theta J(\theta)$

To make a prediction given new x :

Output : $h_\theta(x) = \frac{1}{1+e^{-\theta^T x}}$



Gradient Descent Algorithm

$$J(\theta) = -\frac{1}{m}\left[\sum_{i=1}^{m} y^{(i)} \log\left(h_\theta\left(x^{(i)}\right)\right) + (1 - y^{(i)}) \log\left(1 - h_\theta\left(x^{(i)}\right)\right)\right]$$

Goal: $\min_{\theta} J(\theta)$

Repeat

{

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta)$$

}

$$\boxed{\frac{\partial}{\partial \theta_j} J(\theta) = \frac{1}{m}\sum_{i=1}^{m} (h_\theta\left(x^{(i)}\right) - y^{(i)})\, x_j^{(i)}}$$

Gradient Descent Algorithm

Linear Regression

- Repeat {

$$\theta_j := \theta_j - \alpha \frac{1}{m} \sum_{i=1}^{m} \left(h_\theta\left(x^{(i)}\right) - y^{(i)}\right) x_j^{(i)}$$

}

$$h_\theta(x) = \theta^\top x$$

Logistic Regression

Repeat {

$$\theta_j := \theta_j - \alpha \frac{1}{m} \sum_{i=1}^{m} \left(h_\theta\left(x^{(i)}\right) - y^{(i)}\right) x_j^{(i)}$$

}

$$h_\theta(x) = \frac{1}{1 + e^{-\theta^\top x}}$$

$$h_\theta(x) = \frac{1}{1 + e^{-(\theta_0 + \theta_1 CGPA + \theta_2 IQ)}}$$

Slide credit: Andrew Ng

Application of Logistic Regression
&
Problem Types

Example: Sentiment Analysis – With Engineered features

$x_2=2$ $x_3=1$

It's hokey. There are virtually no surprises , and the writing is second-rate.
So why was it so enjoyable ? For one thing , the cast is
great . Another nice touch is the music I was overcome with the urge to get off
the couch and start dancing . It sucked me in , and it'll do the same to you .

$x_1=3$ $x_5=0$ $x_6=4.19$ $x_4=3$

Sentiment Features

Var	Definition	Value in Fig. 5.2
x_1	count(positive lexicon) $\in$ doc)	3
x_2	count(negative lexicon) $\in$ doc)	2
x_3	$\begin{cases} 1 & \text{if "no"} \in \text{doc} \\ 0 & \text{otherwise} \end{cases}$	1
x_4	count(1st and 2nd pronouns $\in$ doc)	3
x_5	$\begin{cases} 1 & \text{if "!"} \in \text{doc} \\ 0 & \text{otherwise} \end{cases}$	0
x_6	log(word count of doc)	$\ln(66) = 4.19$

Classifying sentiment using logistic regression

Suppose w = $[2.5, -5.0, -1.2, 0.5, 2.0, 0.7]$

b = 0.1

$$
\begin{aligned}
p(+|x) = P(Y=1|x) &= \sigma(w \cdot x + b) \\
&= \sigma([2.5, -5.0, -1.2, 0.5, 2.0, 0.7] \cdot [3, 2, 1, 3, 0, 4.19] + 0.1) \\
&= \sigma(.833) \\
&= 0.70
\end{aligned}
$$

$$
\begin{aligned}
p(-|x) = P(Y=0|x) &= 1 - \sigma(w \cdot x + b) \\
&= 0.30
\end{aligned}
$$

Logistic Regression – Fit a Model – Apply Gradient Descent

CGPA	IQ	IQ	Job Offered
5.5	6.7	100	1
5	7	105	0
8	6	90	1
9	7	105	1
6	8	120	0
7.5	7.3	110	0

$$\theta_0 := \theta_0 - 0.3\frac{1}{6}\sum_{i=1}^{6}\left(h_\theta\left(x^{(i)}\right) - y^{(i)}\right)(1)$$

$$\theta_{CGPA} := \theta_{CGPA} - 0.3\frac{1}{6}\sum_{i=1}^{6}\left(h_\theta\left(x^{(i)}\right) - y^{(i)}\right)x_{CGPA}^{(i)}$$

$$\theta_{IQ} := \theta_{IQ} - 0.3\frac{1}{6}\sum_{i=1}^{6}\left(h_\theta\left(x^{(i)}\right) - y^{(i)}\right)x_{IQ}^{(i)}$$

Hyper parameters:
Learning Rate = 0.3
Initial Weights = (0.5, 0.5,0.5)
Regularization Constant = 0

$\theta^{\mathsf{T}}X$ = 0.5 +0.5 CGPA + 0.5 IQ

$$h_\theta(x) = \frac{1}{1 + e^{-\theta^{\mathsf{T}}x}}$$

$$h_\theta(x) = \frac{1}{1 + e^{-(0.5\ +0.5\ CGPA\ +\ 0.5\ IQ\ )}}$$

Approx. : New weights
θ_0 = 0.4
$\theta_{1=CGPA}$ =-0.4
$\theta_{2=IQ}$ $= -0.6$

Logistic Regression – Inference & Interpretation

CGPA	IQ	IQ	Job Offered
5.5	6.7	100	1
5	7	105	0
8	6	90	1
9	7	105	1
6	8	120	0
7.5	7.3	110	0

Assume : 0.4+0.3CGPA-0.45IQ

Predict the Job offered for a candidate : (5, 6)
h(x) = 0.31
Y-Predicted = 0 / No

Note :
The exponential function of the regression coefficient ($e^{w\text{-}cpga}$) is the odds ratio associated with a one-unit increase in the cgpa.
+ The odd of being offered with job increase by a factor of 1.35 for every unit increase in the CGPA [np.exp(model.params)]

Regularization

Note : This topic is already covered in the module 3 and the implementation remains the same. **Only one points added here w.r.t interpretation for logistic regression**

Ways to Control Overfitting – Interpretation of Hyper parameter

- Regularization

$$Loss(S) = \sum_{i}^{n} Loss(\hat{y_i}, y_i) \quad + \alpha \sum_{j}^{\#Weights} |\theta_j|$$

Note:
The hyperparameter controlling the regularization strength of a Scikit-Learn LogisticRegression model is not alpha (as in other linear models), but its inverse: C. The higher the value of C, the less the model is regularized.

Evaluation of Classifiers

Using Another Example

Evaluation of Classification Model

Confusion Matrix

Mileage (in kmpl)	*Car Price (in cr)*
9.8	High
9.12	Low
9.5	High
10	Low
….	…

		PREDICTED CLASS	
ACTUAL CLASS		Class= Low	Class= High
	Class= Low	a (TP)	b (FN)
	Class= High	c (FP)	d (TN)

$$\text{Accuracy} = \frac{a+d}{a+b+c+d} = \frac{TP+TN}{TP+TN+FP+FN}$$

Unseen Data	
Mileage (in kmpl)	***Car Price (in cr)***
7.5	High
10	Low
….	…..

Most effective when the class distribution is relatively balanced

Model 1

$$\text{CarPrice} = \frac{1}{1+e^{-8.5 + 0.5\,\text{Mileage} - 1.5\,\text{Mileage}^2}}$$

Accuracy: 99%

Model 2

$$\text{CarPrice} = \frac{1}{1+e^{5.5-1.5\,\text{Mileage}}}$$

Accuracy: 50%

Evaluation of Classification Model

Confusion Matrix

		PREDICTED CLASS	
ACTUAL CLASS		Class= Low	Class= High
	Class= Low	0 (TP)	10 (FN)
	Class= High	0 (FP)	990 (TN)

If a model predicts everything to be class NO, accuracy is 990/1000 = 99 %. This is misleading because this trivial model does not detect any class YES example Detecting the rare class is usually more interesting (e.g., frauds, intrusions, defects, etc)

		PREDICTED CLASS	
ACTUAL CLASS		Class= Low	Class= High
	Class= Low	10 (TP)	0 (FN)
	Class= High	500 (FP)	490 (TN)

$$\text{Accuracy} = \frac{a+d}{a+b+c+d} = \frac{TP+TN}{TP+TN+FP+FN}$$

Which model is better?

Model 1

$$\text{CarPrice} = \frac{1}{1+e^{-8.5 + 0.5\,\text{Mileage} - 1.5\,\text{Mileage}^2}}$$

Accuracy: 99%

Model 2

$$\text{CarPrice} = \frac{1}{1+e^{5.5-1.5\,\text{Mileage}}}$$

Accuracy: 50%

Evaluation of Classification Model

Confusion Matrix

		PREDICTED CLASS	
ACTUAL CLASS		Class= Low	Class= High
	Class= Low	a (TP)	b (FN)
	Class= High	c (FP)	d (TN)

$$Accuracy = \frac{TP + TN}{TP + FN + FP + TN}$$

$$ErrorRate = 1 - accuracy$$

$$Precision = Positive\ Predictive\ Value = \frac{TP}{TP + FP}$$

$$Recall = Sensitivity = TP\ Rate = \frac{TP}{TP + FN}$$

$$Specificity = TN\ Rate = \frac{TN}{TN + FP}$$

$$FP\ Rate = \alpha = \frac{FP}{TN + FP} = 1 - specificity$$

$$FN\ Rate = \beta = \frac{FN}{FN + TP} = 1 - sensitivity$$

$$Power = sensitivity = 1 - \beta$$

The F-score (also known as the F1 score or F-measure, combines precision and recall into a single score . F1-score is a better metric when there are imbalanced classes (More on this in upcoming slides)

F-score
= 2 * (precision * recall) / (precision + recall)

Which Classifier is better?

Low Skew Case

T1	PREDICTED CLASS		
ACTUAL CLASS		Class=Yes	Class=No
	Class=Yes	50	50
	Class=No	1	99

$$\text{Precision (p)} = 0.98$$
$$\text{TPR} = \text{Recall (r)} = 0.5$$
$$\text{FPR} = 0.01$$
$$\text{TPR/FPR} = 50$$
$$\text{F} - \text{measure} = 0.66$$

T2	PREDICTED CLASS		
ACTUAL CLASS		Class=Yes	Class=No
	Class=Yes	99	1
	Class=No	10	90

$$\text{Precision (p)} = 0.9$$
$$\text{TPR} = \text{Recall (r)} = 0.99$$
$$\text{FPR} = 0.1$$
$$\text{TPR/FPR} = 9.9$$
$$\text{F} - \text{measure} = 0.94$$

T3	PREDICTED CLASS		
ACTUAL CLASS		Class=Yes	Class=No
	Class=Yes	99	1
	Class=No	1	99

$$\text{Precision (p)} = 0.99$$
$$\text{TPR} = \text{Recall (r)} = 0.99$$
$$\text{FPR} = 0.01$$
$$\text{TPR/FPR} = 99$$
$$\text{F} - \text{measure} = 0.99$$

Which Classifier is better?

Medium Skew Case

T1	PREDICTED CLASS		
ACTUAL CLASS		Class=Yes	Class=No
	Class=Yes	50	50
	Class=No	10	990

$$\text{Precision (p)} = 0.83$$
$$\text{TPR} = \text{Recall (r)} = 0.5$$
$$\text{FPR} = 0.01$$
$$\text{TPR/FPR} = 50$$
$$\text{F} - \text{measure} = 0.62$$

T2	PREDICTED CLASS		
ACTUAL CLASS		Class=Yes	Class=No
	Class=Yes	99	1
	Class=No	100	900

$$\text{Precision (p)} = 0.5$$
$$\text{TPR} = \text{Recall (r)} = 0.99$$
$$\text{FPR} = 0.1$$
$$\text{TPR/FPR} = 9.9$$
$$\text{F} - \text{measure} = 0.66$$

T3	PREDICTED CLASS		
ACTUAL CLASS		Class=Yes	Class=No
	Class=Yes	99	1
	Class=No	10	990

$$\text{Precision (p)} = 0.9$$
$$\text{TPR} = \text{Recall (r)} = 0.99$$
$$\text{FPR} = 0.01$$
$$\text{TPR/FPR} = 99$$
$$\text{F} - \text{measure} = 0.94$$

Which Classifier is better?

High Skew Case

T1	PREDICTED CLASS		
ACTUAL CLASS		Class=Yes	Class=No
	Class=Yes	50	50
	Class=No	100	9900

$$\text{Precision (p)} = 0.3$$
$$\text{TPR} = \text{Recall (r)} = 0.5$$
$$\text{FPR} = 0.01$$
$$\text{TPR/FPR} = 50$$
$$\text{F} - \text{measure} = 0.375$$

T2	PREDICTED CLASS		
ACTUAL CLASS		Class=Yes	Class=No
	Class=Yes	99	1
	Class=No	1000	9000

$$\text{Precision (p)} = 0.09$$
$$\text{TPR} = \text{Recall (r)} = 0.99$$
$$\text{FPR} = 0.1$$
$$\text{TPR/FPR} = 9.9$$
$$\text{F} - \text{measure} = 0.165$$

T3	PREDICTED CLASS		
ACTUAL CLASS		Class=Yes	Class=No
	Class=Yes	99	1
	Class=No	100	9900

$$\text{Precision (p)} = 0.5$$
$$\text{TPR} = \text{Recall (r)} = 0.99$$
$$\text{FPR} = 0.01$$
$$\text{TPR/FPR} = 99$$
$$\text{F} - \text{measure} = 0.66$$

Which Model should you use?

	False Positive Rate	False Negative Rate
Model 1	41%	3%
Model 2	5%	25%

Mistakes have different costs:

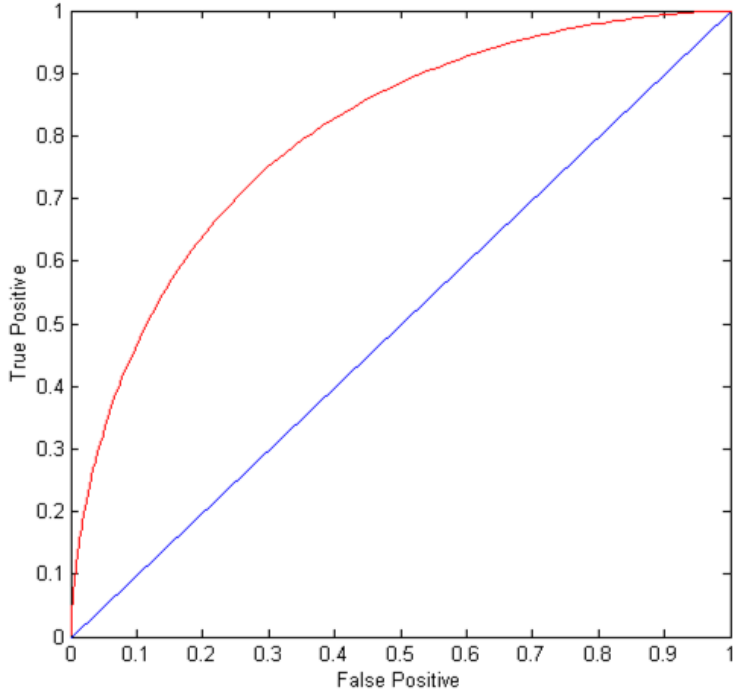
- Disease Screening – LOW FN Rate
- Spam filtering – LOW FP Rate

Conservative vs Aggressive settings:

- The same application might need multiple tradeoffs

ROC (Receiver Operating Characteristic)

- A graphical approach for displaying trade-off between detection rate and false alarm rate
- Developed in 1950s for signal detection theory to analyze noisy signals
- ROC curve plots TPR against FPR
  - Performance of a model represented as a point in an ROC curve

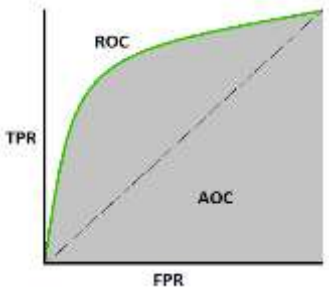


(TPR,FPR):

- (0,0): declare everything to be negative class
- (1,1): declare everything to be positive class
- (1,0): ideal
- Diagonal line:
  - Random guessing
  - Below diagonal line:
    - prediction is opposite of the true class

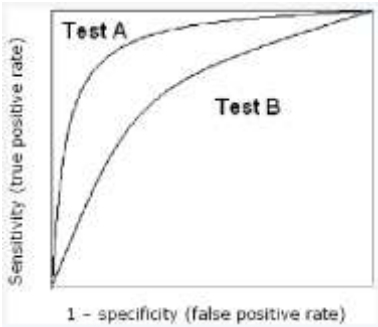
Receiver Operating Characteristic (ROC) Curve

- ROC curve plots TPR and FPR, to graphically represent their trade-off
- AUC represents degree or measure of separability. It tells how much model is capable of distinguishing between classes.
  - Higher the AUC, better the model is at predicting
- Area Under Curve (AUC) of ROC – evaluates model performance on
  - AUC of ROC = 1, for a perfect model
  - AUC of ROC = 0.5, if the model is random
- For model comparison, AUC of ROC should be larger for the model to be superior or better performing



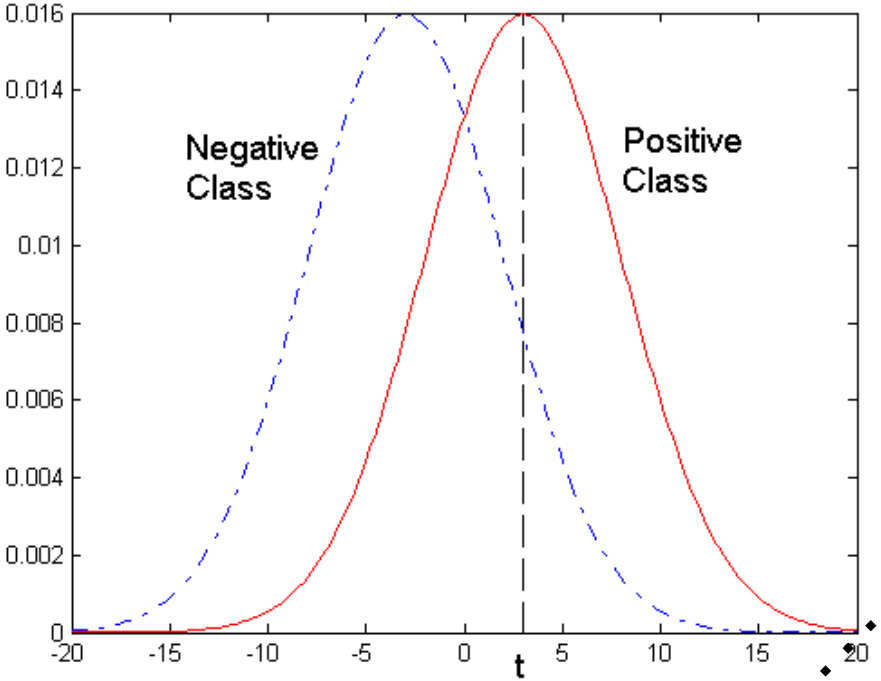
Usage

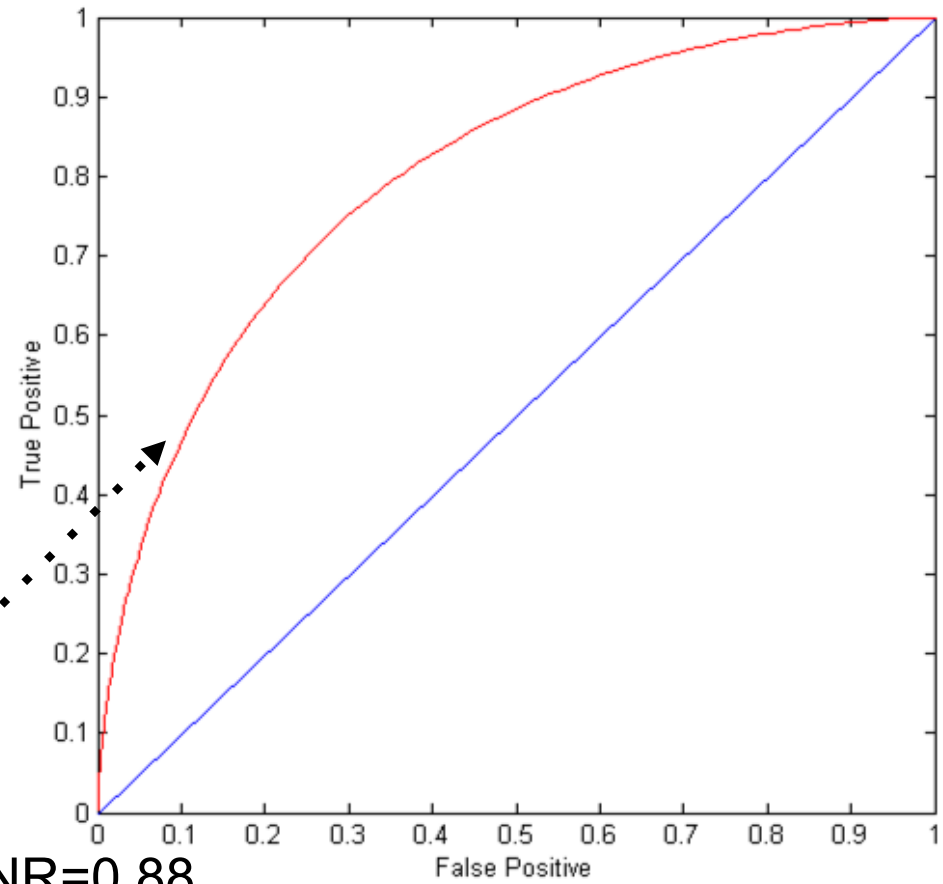
- Threshold selection
- Performance assessment
- Classifier comparison



ROC Curve Example

- 1-dimensional data set containing 2 classes (positive and negative)
- Any points located at x > t is classified as positive





At threshold t:

TPR=0.5, FNR=0.5, FPR=0.12, TNR=0.88

How to Construct an ROC curve

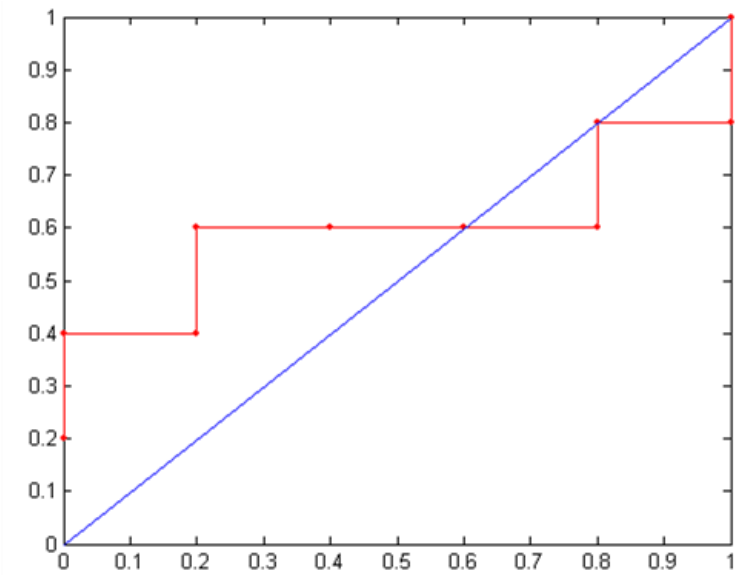
Instance	Score	True Class
1	0.95	+
2	0.93	+
3	0.87	-
4	0.85	-
5	0.85	-
6	0.85	+
7	0.76	-
8	0.53	+
9	0.43	-
10	0.25	+

- Use a classifier that produces a continuous-valued score for each instance
  - The more likely it is for the instance to be in the + class, the higher the score
- Sort the instances in decreasing order according to the score
- Apply a threshold at each unique value of the score
- Count the number of TP, FP, TN, FN at each threshold
  - TPR = TP/(TP+FN)
  - FPR = FP/(FP + TN)

How to construct an ROC curve

	Class	+	-	+	-	-	-	+	-	+	+	
Threshold >=		0.25	0.43	0.53	0.76	0.85	0.85	0.85	0.87	0.93	0.95	1.00
	TP	5	4	4	3	3	3	3	2	2	1	0
	FP	5	5	4	4	3	2	1	1	0	0	0
	TN	0	0	1	1	2	3	4	4	5	5	5
	FN	0	1	1	2	2	2	2	3	3	4	5
→	TPR	1	0.8	0.8	0.6	0.6	0.6	0.6	0.4	0.4	0.2	0
→	FPR	1	1	0.8	0.8	0.6	0.4	0.2	0.2	0	0	0

ROC Curve:



Instance	Score	True Class
1	0.95	+
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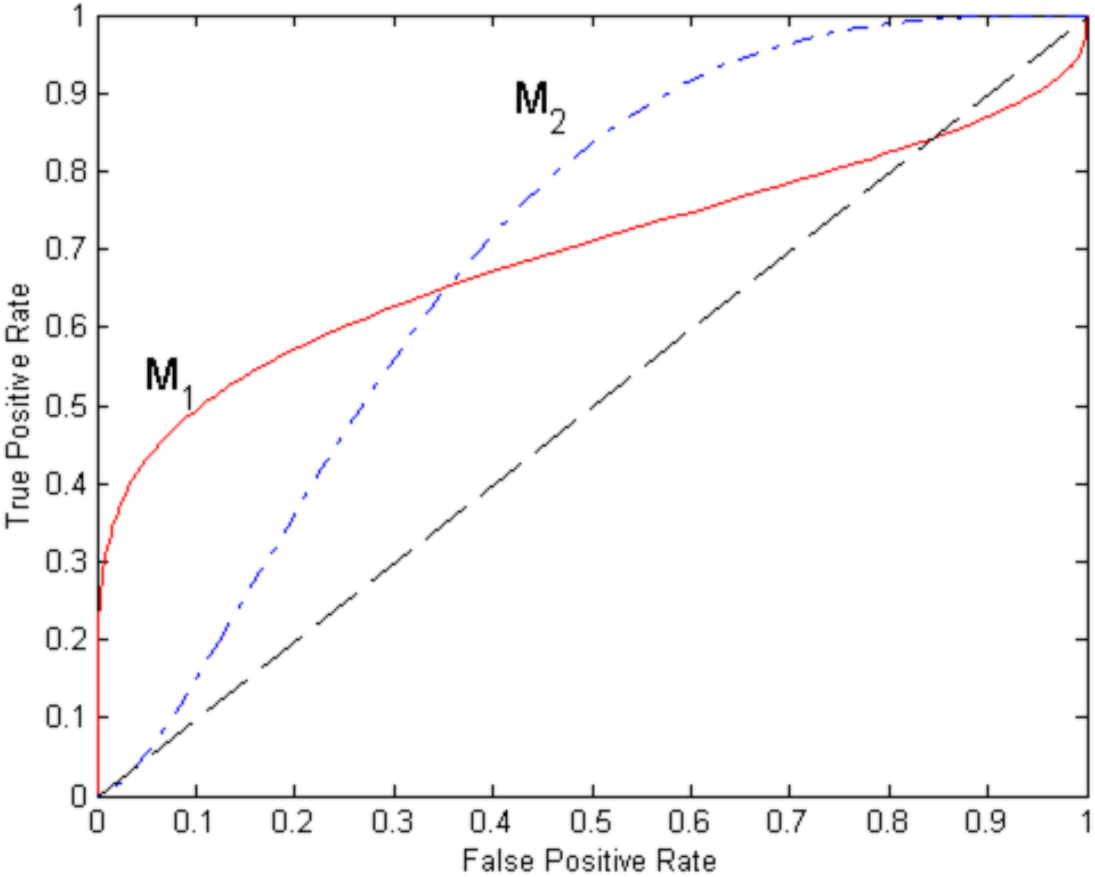
4	0.85	-
5	0.85	-
6	0.85	+
7	0.76	-
8	0.53	+
9	0.43	-
10	0.25	+

Using ROC for Model Comparison

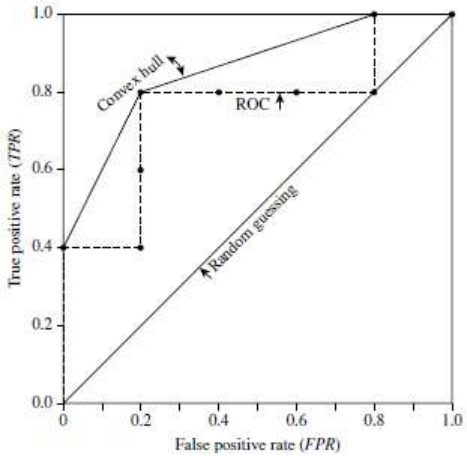


- No model consistently outperforms the other
  - M_1 is better for small FPR
  - M_2 is better for large FPR
- Area Under the ROC curve (AUC)
  - Ideal:
- Area = 1
  - Random guess:
- Area = 0.5

Example

The table below shows the probability value (column 3) returned by a probabilistic classifier for each of the 10 tuples in a test set, sorted by decreasing probability order. The corresponding ROC is given on right hand side.

Tuple #	Class	Prob.	TP	FP	TN	FN	TPR	FPR
1	*P*	0.90	1	0	5	4	0.2	0
2	*P*	0.80	2	0	5	3	0.4	0
3	*N*	0.70	2	1	4	3	0.4	0.2
4	*P*	0.60	3	1	4	2	0.6	0.2
5	*P*	0.55	4	1	4	1	0.8	0.2
6	*N*	0.54	4	2	3	1	0.8	0.4
7	*N*	0.53	4	3	2	1	0.8	0.6
8	*N*	0.51	4	4	1	1	0.8	0.8
9	*P*	0.50	5	4	0	1	1.0	0.8
10	*N*	0.40	5	5	0	0	1.0	1.0



Common Issues in Classifiers

-

BITS Pilani

Sigmoid Function

- The main class of interest is **rare**.
- The data set distribution reflects a significant majority of the negative class (Eg., Job Offered = Yes/1) and a minority positive class (Eg., Job Offered = No/0)

- For Another Example,
  - fraud detection applications, the class of interest (or positive class) is *"fraud,"*
  - *m*edical tests, there may be a rare class, such as *"cancer"*

- **Accuracy might not be a good option** for measuring performance in case of class imbalance problem

Solution to Class Imbalance

- Generate Synthetic Samples
- New samples based on the distances between the point and its nearest neighbors E.g. Synthetic Minority Oversampling Technique, or SMOTE class in sklearn
- Change the performance metric : Use Recall, Precision or ROC curves instead of accuracy
- Try different algorithms : Some algorithms as Support Vector Machines and Tree-Based algorithms may work better with imbalanced classes. We will discuss these post mid term

Logistic Regression –Additional Practice Exercises

CGPA	IQ	IQ	Job Offered
5.5	6.7	100	1
5	7	105	0
8	6	90	1
9	7	105	1
6	8	120	0
7.5	7.3	110	0

Hyper parameters:
Learning Rate = 0.8
Initial Weights = (-0.1, 0.2,-0.5)
Regularization Constant = 10

For this similar problem discussed in class note that the hyper parameters are different

1. Formulate the gradient descent update equations for this problem
2. Repeat the GD for two iterations
3. Find the Loss at every iterations and interpret your observation
4. Using the results of second iteration answer below questions:
   a) Interpret the influence of the CGPA in the response variable
   b) Predict if a new candidates with IQ=5 and CGPA = 9 will be offered job or not?
5. Repeat the steps 2 to 4 by using stochastic gradient descent instead of batch gradient descent for 4 iterations. (Take any random sample from among 6 instances for these 4 iterations)

Evaluation of Classifiers–Additional Practice Exercises

Given below is a confusion matrix for medical data where the class values are yes and no for a class label attribute, cancer. Answer the following questions.

Classes	*yes*	*no*	*Total*	*Recognition (%)*
yes	**90**	**210**	300	30.00
no	**140**	**9560**	9700	98.56
Total	230	9770	10,000	96.40

Confusion matrix for the classes *cancer* = *yes* and *cancer* = *no*.

1. Calculate the Precision , Recall, F-Score, Error-rate, F-Score
2. Brainstorm on the use case / scenarios w.r.t given example, where precision is preferred over recall.
3. Brainstorm on the use case / scenarios w.r.t given example, where recall is preferred over precision.

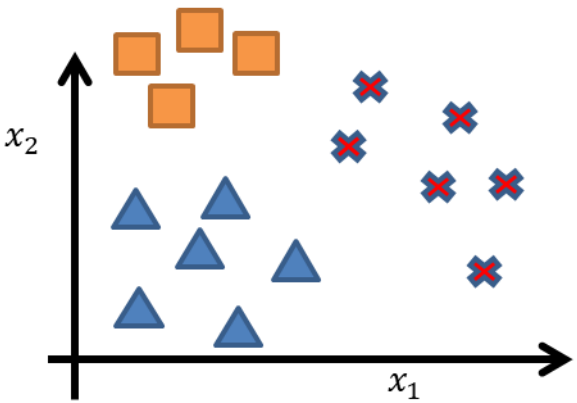
Types of Classification Based on the Output Labels

Types of Classification

Output Labels

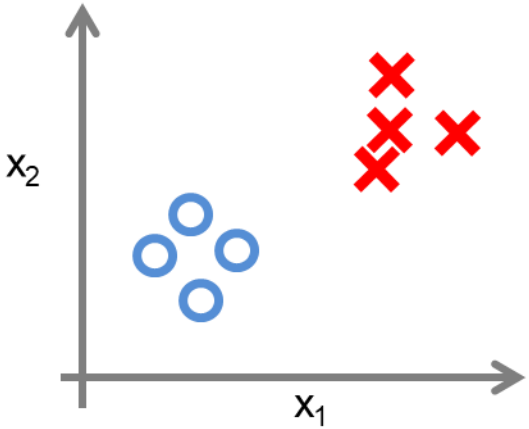
- Target Concept

Multi Class

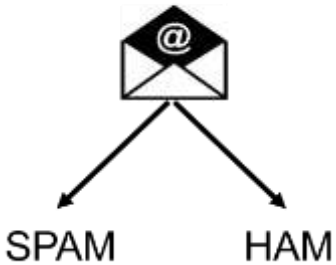


Enjoy Sports

Binary

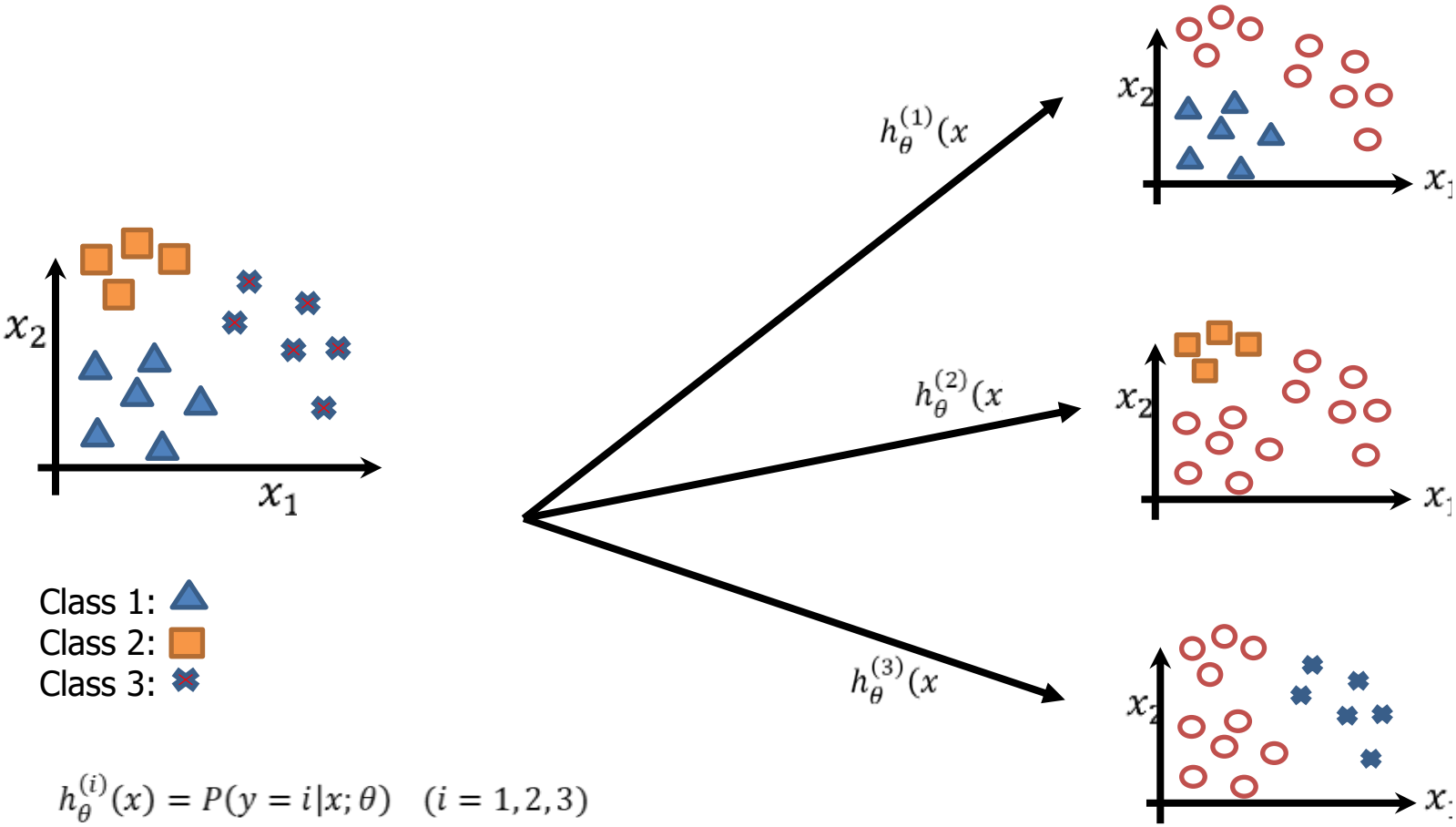


Spam Classifier



Prediction – Multi class Classification

One Vs All Strategy(one-vs-rest)



Class 1: ▲
Class 2: ■
Class 3: ✖

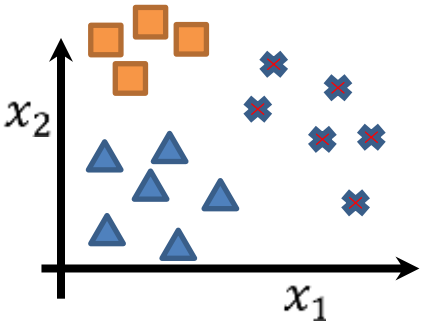
$$h_\theta^{(i)}(x) = P(y = i|x;\theta) \quad (i = 1,2,3)$$

$$\text{For input x Predict}: \ \max_i h_\theta^{(i)}(x)$$

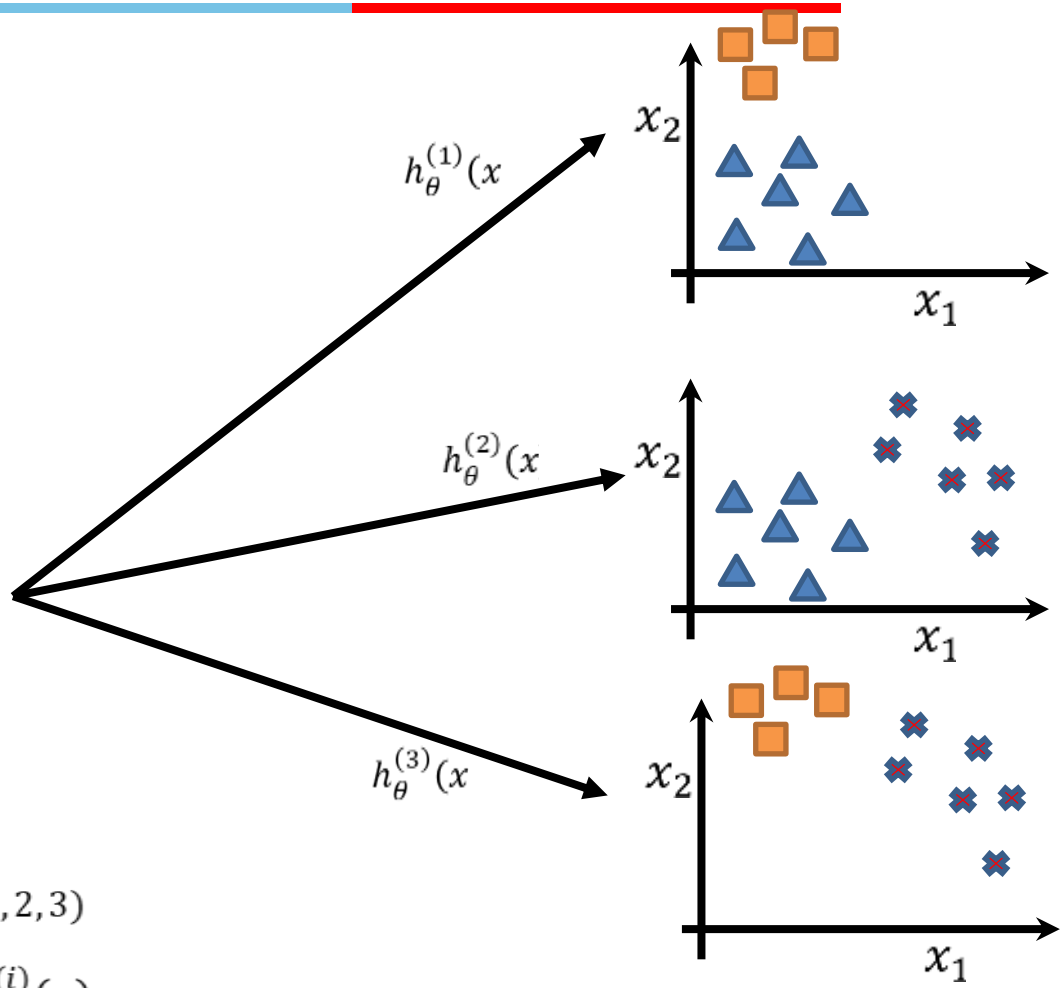
Note: Scikit-Learn detects when you try to use a binary classification algorithm for a multi-class classification task, and it automatically runs OvA (except for SVM classifiers for which it uses OvO)

Prediction – Multi class Classification

One Vs One Strategy



Class 1:
Class 2:
Class 3:



$$h_\theta^{(i)}(x) = P(y = i|x;\theta) \quad (i = 1,2,3)$$

$$\text{For input x Predict}: \ \max_i h_\theta^{(i)}(x)$$

N × (N – 1) / 2 classifiers

Logistic regression (Classification)- Summary

- **Model**

$$h_\theta(x) = P(Y = 1|X_1, X_2, \cdots, X_n) = \frac{1}{1+e^{-\theta^\top x}}$$

- **Cost function**

$$J(\theta) = \frac{1}{m}\sum_{i=1}^{m} \mathrm{Cost}(h_\theta(x^{(i)}), y^{(i)})) \qquad \mathrm{Cost}(h_\theta(x), y) = \begin{cases} -\log(h_\theta(x)) & \text{if } y = 1 \\ -\log(1 - h_\theta(x)) & \text{if } y = 0 \end{cases}$$

- **Learning**

Gradient descent: Repeat $\{\theta_j := \theta_j - \alpha \frac{1}{m}\sum_{i=1}^{m}\left(h_\theta\left(x^{(i)}\right) - y^{(i)}\right) x_j^{(i)}\}$

- **Inference**

$$\hat{Y} = h_\theta(x^{\text{test}}) = \frac{1}{1 + e^{-\theta^\top x^{\text{test}}}}$$

Note:

- σ(t) < 0.5 when t < 0, and σ(t) ≥ 0.5 when t ≥ 0, so a Logistic model predicts 1 if xTθ is positive, and 0 if it is negative
- logit(p) = log(p / (1 - p)), is the inverse of the logistic function. Indeed, if you compute the logit of the estimated probability p, you will find that the result is t. The logit is also called the log-odds

Thank you !

Required Reading for completed session :

T1 - Chapter # 6 (Tom M. Mitchell, Machine Learning)

R1 – Chapter # 3,#4 (Christopher M. Bhisop, Pattern Recognition & Machine Learning) & Refresh your MFDS course basics

Next Session Plan :

Module 5 – Decision Tree Classifier